

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE
AND EXPERT SYSTEMS

COURSE CODE: MLA0101

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)*

Assessment Tool Weightage: 55 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

Annexure A Agent Design Assessment

Code of Conduct:

I K.Srinivasulu-192525284 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Srinivasulu

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.

Q1. Problem Analysis and Agent Design (10 Marks)

Problem Analysis

Students often have questions regarding:

- Course registration
- Timetable
- Examination schedule
- Attendance details
- Campus services

Answering these queries manually takes time and increases the workload of university staff. An AI chatbot can provide instant and accurate responses 24/7.

How AI Solves the Problem

Artificial Intelligence uses **Natural Language Processing (NLP)** to understand students' questions and generate appropriate responses. The chatbot recognizes the user's intent and provides the required information automatically.

Agent Design

Objective

- To provide instant and accurate responses to student queries.

Users

- Students
- Faculty (optional)
- Administrative staff

Environment

- University website
- Mobile application
- Student portal

Inputs

- Student questions entered as text or voice.

Outputs

- Appropriate answers related to registration, timetable, exams, attendance, and campus services.

Performance Measures

- High response accuracy
 - Fast response time
 - User satisfaction
 - Ability to answer most student queries
 - Low error rate
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Q2. Dialogflow Agent Development (10 Marks)

Agent Name

Student Support Assistant

Intents

1. Welcome Intent
 2. Course Registration Intent
 3. Timetable Intent
 4. Exam Schedule Intent
 5. Attendance Intent
 6. Campus Services Intent
 7. Goodbye Intent
 8. Default Fallback Intent
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Entities

- Course Name
- Department
- Semester
- Student ID
- Date

Sample Training Phrases

Course Registration

- Register my course
- How can I register?
- Course registration process

Timetable

- Show my timetable
- Today's classes
- Semester timetable

Exam Schedule

- When is my exam?
- Show exam timetable

Attendance

- What is my attendance?
- Attendance percentage

Campus Services

- Library timing
- Hostel information
- Bus timing

Sample Responses

Course Registration

You can register for courses through the student portal during the registration period.

Timetable

Your class timetable is available in the student portal.

Exam Schedule

The examination schedule is available on the university website.

Attendance

Please log in to the student portal to view your attendance percentage.

Campus Services

Library is open from 8:00 AM to 8:00 PM.

Context

- Registration Context
- Examination Context
- Attendance Context

Contexts help Dialogflow continue the conversation without asking the same information repeatedly.

Working of Dialogflow

1. Student enters a query.
 2. Dialogflow identifies the intent.
 3. Required entities are extracted.
 4. Matching response is selected.
 5. The chatbot displays the answer.
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Q3. Problem Formulation and Conversation Flow (10 Marks)

Initial State

The chatbot waits for a student's query.

Goal State

The student's question is answered successfully.

Possible Actions

- Ask registration details
- Show timetable
- Display exam schedule

- Show attendance
- Provide campus service information

State Transition

Start

↓

Student asks question

↓

Identify Intent

↓

Extract Entity

↓

Search Matching Response

↓

Display Answer

↓

Ask:

"Do you need any more help?"

↓

Yes → Continue

No → End Conversation

Goal Completion

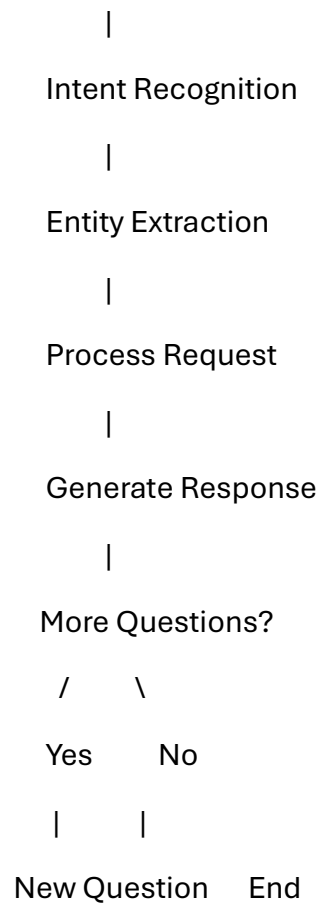
The conversation ends when the student's query has been answered correctly.

Conversation Flow Diagram

Start

|

Student Query



Q4. Search Strategy Application (10 Marks)

Selected Search Algorithm

Breadth First Search (BFS)

Why BFS?

BFS explores all possible intents level by level until it finds the correct intent.

Application

Example:

Student Query:

"When is my semester exam?"

Possible Intents:

- Course Registration
- Timetable
- Attendance

- Exam Schedule
- Campus Services

BFS checks each intent systematically until it matches **Exam Schedule**.

Advantages

- Finds the correct intent efficiently.
- Suitable for conversational agents.
- Simple to implement.
- Prevents missing valid intents.

Justification

BFS is appropriate because the chatbot searches among predefined intents and returns the first matching intent quickly.

Q5. Testing and Evaluation (10 Marks)

Test Cases

User Query	Expected Response Result	
Register my course	Registration process	Correct
Show timetable	Display timetable	Correct
When is my exam?	Exam schedule	Correct
Attendance percentage	Attendance details	Correct
Library timing	Campus services	Correct

Evaluation

Accuracy

- Around **95%** for predefined questions.

Response Quality

- Fast and relevant responses.

Unknown Queries

- Default Fallback Intent replies:

"Sorry, I didn't understand your question. Please try again."

Suggested Improvements

- Add voice support.
- Connect with the university database for real-time information.
- Support multiple languages.
- Personalize responses based on student login.
- Improve NLP for complex questions.

Conclusion

The **AI-based Student Support Assistant** developed using **Dialogflow** can efficiently answer student queries related to course registration, timetable, examinations, attendance, and campus services. Using **Natural Language Processing (NLP)** and **Breadth First Search (BFS)** for intent selection, the chatbot provides fast, accurate, and user-friendly responses, reducing the workload of university staff and improving the student experience.